CFFormer: A Cross-Fusion Transformer Framework for the Semantic Segmentation of Multisource Remote Sensing Images

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***Abstract*— Multisource remote sensing images (RSIs) can capture the complementary information of ground objects for use in semantic segmentation. However, there can be inconsistency and interference noise among the multimodal data from different sensors. Therefore, it is a challenge to effectively reduce the differences and noise between the different modalities and fully utilize their complementary features. In this article, we propose a universal cross-fusion transformer framework (CFFormer) for the semantic segmentation of multisource RSIs, adopting a parallel dual-stream structure to extract features separately from the different modalities. We introduce a feature correction module (FCM) that corrects the features of the current modality by combining features from the other modalities in both the spatial and channel dimensions. In the feature fusion module (FFM), we employ a multihead cross-attention mechanism to interact globally and fuse features from the different modalities, enabling the comprehensive utilization of the complementary information in multisource RSIs. Finally, comparative experiments demonstrate that the proposed CFFormer framework not only achieves state-of-the-art (SOTA) accuracy but also exhibits outstanding robustness when compared to the current advanced networks for semantic segmentation of multisource RSIs. Specifically, CFFormer achieves a mean intersection over union (mIoU) of 58% and an overall accuracy (OA) of 85.35% on the WHU-OPT-SAR dataset, outperforming the second-ranked network by 4.71% and 1.74%, respectively. On the Vaihingen and Potsdam datasets, CFFormer also achieves the best results, with mIoU and OA values of 84.31%/91.88% and 88.62%/92.64%, respectively. The source code is available at https://github.com/masurq/CFFormer.**

***Index Terms*— Feature correction module (FCM), feature fusion module (FFM), multisource remote sensing images (RSIs), semantic segmentation, vision transformer.**

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I. Introduction

SEMANTIC segmentation of remote sensing images (RSIs) provides support for various RSI application scenarios, such as land-cover classification [1], [2], [3] land planning [4], urban planning [5], [6], [7], and road extraction [8], [9], and has thus become a research hotspot in the field of remote sensing data processing. Among the different data sources, optical RSIs have been the most widely applied, due to their high spatial resolution, rich spectral information, ease of interpretation and understanding, and relatively low acquisition costs. However, as remote sensing technology continues to develop, optical images present challenges in semantic segmentation tasks in complex scenes. For example, optical RSIs are plagued by clouds or adverse weather, which greatly reduces the segmentation accuracy [10]. As a result, it has been found that individual remote sensing data sources are usually insufficient to meet the requirements of the different tasks [11].

As the acquisition of multisource remote sensing data becomes increasingly accessible [12], [13], the comprehensive utilization of the complementary information from optical images and other modal images has become one of the solutions to this problem [14]. For example, digital surface models (DSMs) can provide elevation information for objects on the Earth's surface [15], while synthetic aperture radar (SAR) can offer rich amplitude and phase geometric information independent of illumination and weather conditions [16], [17], [18]. Furthermore, some objects may display comparable interclass appearances in spectral information, as depicted in Fig. 1(a), where patches of villages and cities are challenging to distinguish in optical images while they are distinctly discernible in SAR images. A similar situation can be observed in Fig. 1(b), where trees and low vegetation appear similar in the optical images, but the distinction between them is evident in the DSM images, due to their elevation differences. Therefore, leveraging the complementary information from multisource RSIs to address the constraints of single-source images and enhance the stability and robustness of segmentation is an important research topic.

With the continuous rise of deep learning, research on the semantic segmentation of multisource RSIs based on

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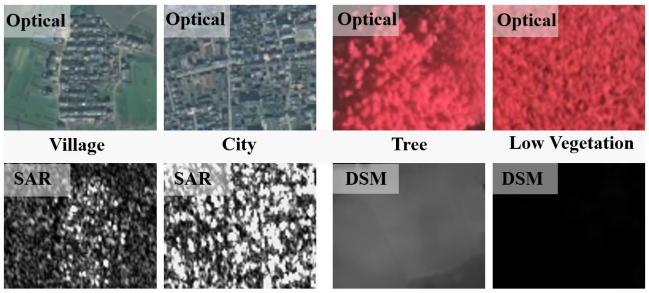


Fig. 1. Examples of the complementarity between RSIs.

this technology has gained more and more attention. The early deep learning methods typically extracted pixel values from the different modalities directly and input them into a multilayer perceptron (MLP) for pixel-level classification. The final prediction results were then obtained by aggregating the classification results [19], [20], [21]. Although these methods are more powerful than the traditional feature engineering techniques, they overlook the contextual relationships of the spatial regions around pixels. To tackle this issue, subsequent deep learning methods introduced convolutional neural networks (CNNs) [22], [23]. These methods extract features from local regions of the image via the CNN and employ sophisticated feature fusion strategies to integrate the complementary information between different modalities, thereby enhancing the accuracy of the models. For instance, V_FuseNet [24] explores the classification performance of optical and DSM data under early fusion and late fusion; the dense connection and fully convolutional network (DFCN) [25] uses DSM data as supplementary information, incorporating a multimodal network combining the ideas of dense connections and a fully convolutional network (FCN), and widening the network using multiscale filters; and the dense-attention-similarity-fusion network (DASFNet) [26] employs cosine similarity along with a gating mechanism to fuse infrared-red-green optical data and DSM data and take advantage of the complementarity of each data source. Using optical data and SAR data, MBFNet [27] establishes the second-order association between optical features and SAR features through bilinear pooling operations; multimodal-cross attention network (MCANet) was proposed [10] based on a cross-attention mechanism and an open-source optical-SAR land-use classification dataset; and the cross fusion network (CFNet) [28] introduces a cross-gate module with bidirectional information flow to fully exploit the complementary information from different modalities. However, these methods only use a relatively simple global multimodal interaction strategy, and cannot be well generalized to different modal combinations.

In addition, the constraints inherent in the convolutional operations of CNNs have made it progressively difficult to achieve substantial advancements. Consequently, a growing number of researchers are turning their focus to transformer-based approaches. These methods harness innovative frameworks to bolster interpretive precision. The transformer architecture [29] is a more powerful global feature extraction network. It converts the 2-D image-based task into a 1-D sequence-based task and extracts the global contextual information through the advantage of a self-attention mechanism, achieving state-of-the-art (SOTA) results in basic visual tasks such as object detection [30], image classification [31], and semantic segmentation [32]. Driven by this, many researchers in the remote sensing community have applied the transformer architecture to semantic segmentation tasks in single-source RSIs [33], [34], [35], [36]. However, for multisource RSIs, most of the semantic segmentation networks are still based on CNN architectures, and these convolutional filters struggle to establish long-range correlations in pixel-rich multimodal images, limiting the performance of the models. Moreover, due to the differences between the various remote sensing sensor modalities, there is often significant inconsistency in the representation of multimodal data [37], [38]. Meanwhile, different types of noise can also lead to mutual interference between multimodal data. In this regard, one can leverage the transformer's capability to model long-range correlations to implement a global multimodal interaction strategy, so as to better mine the potential of cross-modal complementary features.

In this article, to address the challenge of insufficient utilization of complementary information in multimodal data and the difficulty in establishing long-range correlations between multimodal images using CNN, we propose a novel cross-fusion transformer framework (CFFormer), aiming to perform semantic segmentation by jointly leveraging optical remote sensing data and other multimodal remote sensing data (e.g., SAR or DSM data). CFFormer employs a dual-stream architecture based on a transformer model to extract features from the optical modality and other modalities separately. To fully exploit the extracted multimodal features, we have designed a feature correction module (FCM) and a feature fusion module (FFM). The FCM corrects the local spatial features and global semantic features of each modality by using the spatial and channel correlations between different modalities. This correction aims to alleviate the differences and interference noise between the two modalities, directing their attention toward each other's complementary information. The FFM further enhances and fuses the two sets of corrected modal features. By embedding the newly developed efficient cross-attention mechanism, the FFM captures the interactions between different modalities at a global scale, contributing to the full utilization of the complementary information from multimodal data. The highlights of this article are as follows.

1) CFFormer is proposed as a multimodal framework for the semantic segmentation of RSIs, which interacts and fuses the information of multisource RSIs from the perspectives of feature map sequences through the FCM and FFM.

2) An efficient multihead cross-attention mechanism based on a self-attention mechanism is proposed. It aims to fully capture the complex interactions between different modalities. Moreover, introducing additional convolutions reduces the computational complexity and memory requirement.

3) An efficient FCM is proposed to adaptively refine and filter out discrepancies and noise interference between various modalities through spatial and channel correlations of different

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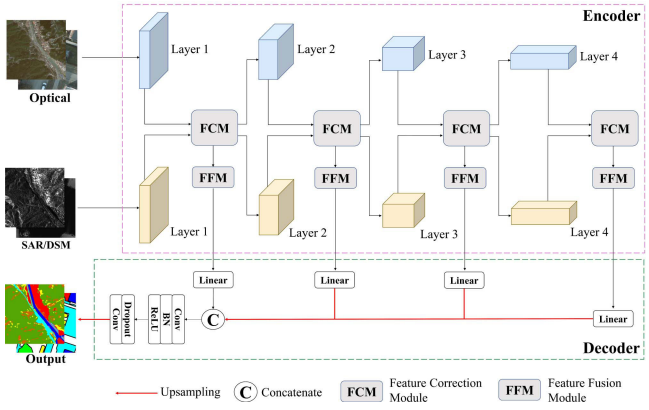


Fig. 2. CFFormer framework. The input to our model consists of two branches, one with an optical image and the other with an SAR or DSM image. After passing through the four-layer encoder of Biformer, the output of each layer in the dual-branch is processed by the FCM module for feature correction and the FFM module for feature fusion. The output of the encoder is upsampled by the decoder to generate a segmentation map.

modalities. Our FCM has enhanced the extraction and fusion of multimodal features by using cross-modal complementary information.

4) Through extensive experimentation with multimodal datasets (ISPRS Vaihingen, Potsdam, and WHU-OPT-SAR), the proposed framework has been shown to be highly capable of semantic segmentation of not only optical and SAR but also optical and DSM data.

The remainder of this article is structured as follows. Section II presents the overall framework of CFFormer, with a full description of each component. The extensive experimental results and thorough analyses are provided in Section III. In Section III-C, the effectiveness and principles of the proposed method are discussed through visualization. Finally, Section IV provides a summary of our work and possible directions for future research.

II. Methodology

A. Proposed CFFormer Framework

We propose a novel network framework based on a transformer model, which uses the FCM and FFM to facilitate the fusion of heterogeneous data sources and achieve more accurate semantic segmentation. The algorithmic framework of this article is shown in Fig. 2. In detail, the proposed approach relies on the classical encoder–decoder architecture, where the encoder incorporates feature extraction networks without weight sharing: the FCM for filtering diverse modal noise and differences, and the FFM for enhancing the information interaction and fusion. The decoder part aggregates the multiscale features to generate the final result. Other common methods such as ResNet can be employed as an alternative for the feature extraction network.

B. Encoder

The encoder comprises three components: feature extraction, feature correction, and feature fusion. Two pretrained transformer-based BiFormer networks [39] are used as the backbones, extracting features from the multimodal remote sensing data in a parallel manner. BiFormer is composed of four stages, each of which downsamples the feature maps by a factor of 2. In the subsequent correction and fusion processes, we make full use of the downsampled features in each stage to better capture information in the image through the acquired multilevel and multiscale feature representations. The FCM comprises spatial-wise and channel-wise feature corrections, leveraging the correlations between the different modalities in both the spatial and channel dimensions to rectify their respective feature information. The FFM further deepens and fuses the corrected multimodal features through a cross-attention mechanism, globally enhancing the information interaction between each pixel. The details of each module are as follow:

1) Feature Correction Module: Due to the differences in the physical principles and measurement methods of the different sensors in data acquisition, different modalities often show significant appearance disparities [37], [38], and are also influenced by various types of noise. However, neighboring regions in the scene will have spatial and geometric consistency across the different modalities. Meanwhile, the semantic structures within the scene will be preserved, to some extent, in the different modalities, as exemplified by buildings, which exhibit clear outlines in optical images, high backscattering intensity in SAR images, and large elevation values in DSM images. This spatial consistency and semantic association enable information from the different modalities to be locally spatially related and globally semantically related. Therefore, we propose the FCM, which includes spatial correction and channel correction. The FCM adaptively corrects and filters the differences and interference noise between the different modalities through spatial correlation and channel correlation, which can focus more on the complementary information between each modality, achieving better multimodal feature extraction and fusion. The order of the feature correction affects the accuracy of the proposed model, as explained in Section III-D2. In this section, the order of spatial-wise correction followed by channel-wise correction is used to introduce the specific structure (see Fig. 3).

a) Spatial-wise feature correction: As shown in the left part of Fig. 3, we denote an optical image as $\mathbf{OPT}_{\text{in}} \in \mathbb{R}^{H\times W\times C}$ and a corresponding SAR image as $\mathbf{SAR}_{\text{in}} \in \mathbb{R}^{H\times W\times C}$, where SAR can be replaced with other complementary modalities such as DSM. We first concatenate the different modalities and then perform layer-by-layer nonlinear transformation and feature learning on the concatenated data using an MLP layer to obtain a feature map $\mathbf{F}^S \in \mathbb{R}^{H\times W\times 2}$. The MLP layer consists of two linear layers and a rectified linear unit (ReLU) function, as specified in (1) and (2). Next, the output range of $\mathbf{F}^S$ is changed to [0, 1] using the sigmoid function, and it is further split into two spatial weight maps $\mathbf{W}^S_{\text{OPT}} \in \mathbb{R}^{H\times W}$ and $\mathbf{W}^S_{\text{SAR}} \in \mathbb{R}^{H\times W}$, as shown in (3)

$$\mathbf{F}^S = \text{MLP}(2C, 2)(\text{Concat}(\mathbf{OPT}_{\text{in}}, \mathbf{SAR}_{\text{in}})) \quad (1)$$

$$\text{MLP} = \text{Linear}(C_{\text{mid}}, C_{\text{out}})(\text{ReLU}(\text{Linear}(C_{\text{in}}, C_{\text{mid}})(\cdot))) \quad (2)$$

$$\mathbf{W}^S_{\text{OPT}}, \mathbf{W}^S_{\text{SAR}} = \text{Split}\left(\sigma\left(\mathbf{F}^S\right)\right) \quad (3)$$

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where $\mathrm{Linear}(C_{\mathrm{in}}, C_{\mathrm{mid}})(\cdot)$ represents a linear layer with input channel C_{in} and output channel C_{mid}; $\mathrm{ReLU}(\cdot)$ represents the ReLU function; and $\sigma(\cdot)$ represents the sigmoid function.

The spatial-wise correction offset of the input $\mathbf{OPT}_{\mathrm{in}}$ is obtained by element-wise multiplication of the acquired spatial weight map $\mathbf{W}^S_{\mathrm{SAR}}\mathrm{D}$ with the corresponding input $\mathbf{SAR}_{\mathrm{in}}$. This is then added to $\mathbf{OPT}_{\mathrm{in}}$, resulting in the spatial-wise correction feature map $\mathbf{OPT}^S_{\mathrm{corr}}$. Furthermore, we introduce a trainable parameter α $(0 \le \alpha \le 1)$ to control the dynamic weighting of the relative importance of the spatial-wise correction and channel-wise correction. During the model training process, this parameter is automatically adjusted based on the contributions of the two correction offsets to the segmentation accuracy, so as to dynamically learn the optimal fusion ratio for the two sets of correction features, which enhances the adaptability and robustness of the model [40]. The correction process can be expressed through the following equation:

$$\begin{cases} \mathbf{OPT}^S_{\mathrm{corr}} = \mathbf{OPT}_{\mathrm{in}} + \alpha \cdot \mathbf{W}^S_{\mathrm{SAR}} \cdot \mathbf{SAR}_{\mathrm{in}} \\ \mathbf{SAR}^S_{\mathrm{corr}} = \mathbf{SAR}_{\mathrm{in}} + \alpha \cdot \mathbf{W}^S_{\mathrm{OPT}} \cdot \mathbf{OPT}_{\mathrm{in}} \end{cases} \tag{4}$$

where $\mathbf{OPT}^S_{\mathrm{corr}}$ and $\mathbf{SAR}^S_{\mathrm{corr}}$ represent the optical and SAR feature maps after spatial-wise correction, respectively; $\mathbf{W}^S_{\mathrm{SAR}}$ and $\mathbf{W}^S_{\mathrm{OPT}}$ represent the weights of the spatial-wise correction offsets; and α is a learnable parameter in the spatial-wise correction stage.

b) Channel-wise feature correction: After calibrating the local information of the different modalities using the spatial-wise FCM, we further introduce the channel-wise FCM for global calibration. As shown in the right half of Fig. 3, the features $\mathbf{OPT}^S_{\mathrm{corr}}$ and $\mathbf{SAR}^S_{\mathrm{corr}}$, which have undergone spatial-wise feature correction, are first concatenated to obtain $\mathbf{X}^C \in \mathbb{R}^{H\times W\times 2C}$. For complex multimodal features, a comprehensive pooling strategy is adopted, including global average pooling [41] to capture the overall average information, global max pooling [42] to capture the salient feature information, and global standard deviation pooling to focus on the feature variation and distribution information. This combination of multiple pooling operations can provide a more comprehensive and rich feature representation, enabling the model to adapt to different types and distributions of data. Subsequently, the global feature vectors obtained after these three pooling operations are concatenated to form a feature map $\mathbf{Y}^C$ of size $\mathbb{R}^{6C}$. Next, the MLP layers are applied to obtain $\mathbf{F}^C \in \mathbb{R}^{2C}$, and a sigmoid activation function and slicing operations are used to obtain the channel weight maps $\mathbf{W}^C_{\mathrm{OPT}} \in \mathbb{R}^C$ and $\mathbf{W}^C_{\mathrm{SAR}} \in \mathbb{R}^C$ that contains the global feature information. The above process can be summarized by the following equation:

$$\begin{cases} \mathbf{X}^C = \mathrm{Concat}\left(\mathbf{OPT}^S_{\mathrm{corr}}, \mathbf{SAR}^S_{\mathrm{corr}}\right) \\ \mathbf{Y}^C = \mathrm{Concat}\left(\mathrm{Avg}\left(\mathbf{X}^C\right)\mathrm{Max}\left(\mathbf{X}^C\right)\mathrm{Std}\left(\mathbf{X}^C\right)\right) \\ \mathbf{F}^C = \mathrm{MLP}(6C, 2C)\left(\mathbf{Y}^C\right) \\ \mathbf{W}^C_{\mathrm{OPT}}, \mathbf{W}^C_{\mathrm{SAR}} = \mathrm{Split}\left(\sigma\left(\mathbf{F}^C\right)\right) \end{cases} \tag{5}$$

where the MLP is shown in (2) and contains two linear layers and a ReLU activation function; $\mathrm{Avg}(\cdot), \mathrm{Max}(\cdot)$, and refer to global average pooling, global max pooling, and global standard deviation pooling, respectively; and $\mathrm{Std}(\cdot)$ is a sigmoid function.

Similar to the spatial-wise feature correction, the channel-wise feature correction can be expressed as follows:

$$\begin{cases} \mathbf{OPT}^{S=C}_{\mathrm{corr}} = \mathbf{OPT}^S_{\mathrm{corr}} + (1-\alpha) \cdot \mathbf{W}^C_{\mathrm{SAR}} \cdot \mathbf{SAR}^S_{\mathrm{corr}} \\ \mathbf{SAR}^{S=C}_{\mathrm{corr}} = \mathbf{SAR}^S_{\mathrm{corr}} + (1-\alpha) \cdot \mathbf{W}^C_{\mathrm{OPT}} \cdot \mathbf{OPT}^S_{\mathrm{corr}} \end{cases} \tag{6}$$

where $\mathbf{OPT}^S_{\mathrm{corr}}$ and $\mathbf{SAR}^S_{\mathrm{corr}}$ refer to the optical and SAR feature maps after spatial-wise correction, respectively; $\mathbf{OPT}^{S=C}_{\mathrm{corr}}$ and $\mathbf{SAR}^{S=C}_{\mathrm{corr}}$ represent the optical and SAR feature maps after spatial-wise correction and channel-wise correction, respectively; $1-\alpha$ denotes the learnable parameter in the channel-wise correction stage, and α in (6) is the same as in (4), controlling the relative importance of the spatial-wise correction stage and the channel-wise correction stage; $\mathbf{W}^C_{\mathrm{SAR}}$ and $\mathbf{W}^C_{\mathrm{OPT}}$ represent the weights of the channel-wise correction offsets.

2) Feature Fusion Module: Making full use of the complementarity between different source features is crucial for multisource semantic segmentation, and the pivotal aspect of this process involves effectively modeling the complex interactions between the different modalities. Currently, deep learning-based methods for multimodal information fusion are typically classified into summation-based methods [24], [43], concatenation-based methods [44], and those based on cross-attention mechanisms [10]. However, the first two approaches not only lead to feature redundancy, making the fusion results prone to local optima and interference noise, but also fail to tap into the complex interactions between the different modalities, significantly reducing the algorithm performance [37]. The methods based on cross-attention mechanisms can not only alleviate this problem by selecting effective information from the different modalities and eliminating redundancy, but also enable global information interaction, which fully captures the complex nonlinear relationships between different modal data. However, this approach is usually complex and computationally demanding.

To address the issues above, we propose the FFM module. This module introduces a cross-attention module to select valid information, realize global information interaction, and fully capture the complex nonlinear relationship from different modal data. In addition, we have introduced a selectable parameter that allows the model to perform operations at different scales. Specifically, we downsample the input sequence, using an additional convolutional layer to shorten the sequence length during processing. This can fully capture the global context information and reduce the computational cost, solve the information redundancy caused by applying the multihead attention mechanism for high-resolution feature maps, and enable the model to focus on the important features better. We also implement a novel multihead attention calculation method, which optimizes the computational dimension by reordering the query-key-value calculation order of the multihead attention mechanism, thereby reducing the computational burden. The specific structural design is illustrated in Fig. 4.

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Assuming that the postcorrection inputs for the two modalities are denoted as $\mathbf{OPT}_{\text{corr}}^{S=C} \in \mathbb{R}^{H\times W\times C}$ and $\mathbf{SAR}_{\text{corr}}^{s=C} \in \mathbb{R}^{H\times W\times C}$, they are first flattened into feature sequences of size $\mathbb{R}^{N\times C}$, where $N = H \times W$. Subsequently, a cross-attention mechanism is employed for global information exchange between the two feature sequences. The cross-attention mechanism in the proposed framework is based on the classic self-attention mechanism [29], with the calculation formula for the self-attention mechanism as follows:

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d_{\text{head}}}}\right)\mathbf{V} \tag{7}$$

where $\mathbf{O} \in \mathbb{R}^{N\times C}$, $\mathbf{K} \in \mathbb{R}^{N\times C}$, and $\mathbf{V} \in \mathbb{R}^{N\times M}$ are all generated from the input feature sequences through linear layers, and $N = H \times W$ refers to the length of the input sequences. C is the dimension of sequences $\mathbf{Q}$ and $\mathbf{K}$, while M is the dimension of sequence $\mathbf{V}$. In the proposed method, C is equal to M, and thus for notation simplicity, only C is used. In addition, d_{head} represents the dimension of the feature vectors in each attention head when employing the multihead attention strategy. The computational complexity of the process shown in (7) is $O(N^2)$, which increases quadratically with the length of the sequence. However, for RSIs with large-scale characteristics, this quadratic growth in computational complexity is intolerable. To solve this problem, the sequence reduction method proposed by Xie et al. [45] is employed. Downsampling along the dimension of the sequence length by using a convolution sliding along that dimension is achieved, resulting in a smaller key vector $\mathbf{K} \in \mathbb{R}^{(N/R)\times C}$ and value vector $\mathbf{V} \in \mathbb{R}^{(N/R)\times C}$, where R represents the downsampling factor. Consequently, the matrix multiplication complexity is reduced from $O(N^2)$ to $O((N^2/R))$, which effectively lowers the overall computational complexity and memory usage. To extend the model's expressive capacity and make it more suitable for handling complex input relationships and patterns, a multihead cross-attention mechanism is further introduced. In detail, we first obtain vectors $\mathbf{Q}_{\text{OPT}}^i \in \mathbb{R}^{N\times d_{\text{head}}}$, $\mathbf{K}_{\text{OPT}}^i \in \mathbb{R}^{(N/R)\times d_{\text{head}}}$, $\mathbf{V}_{\text{OPT}}^i \in \mathbb{R}^{(N/R)\times d_{\text{head}}}$, $\mathbf{Q}_{\text{SAR}}^i \in \mathbb{R}^{N\times d_{\text{head}}}$, $\mathbf{K}_{\text{SAR}}^i \in \mathbb{R}^{(N/R)\times d_{\text{head}}}$, and $\mathbf{V}_{\text{SAR}}^i \in \mathbb{R}^{(N/R)\times d_{\text{head}}}$ for each head of the optical and SAR modalities, where i represents the attention head ($1 \le i \le$ heads, where heads is the number of attention heads), and d_{head} represents the dimension of the feature vectors in each attention head. The attention matrices are then calculated for the optical and SAR modalities through the following equation:

$$\begin{cases} \mathbf{S}_{\text{OPT}}^i = \text{Softmax}\left(\dfrac{\mathbf{Q}_{\text{OPT}}^i \mathbf{K}_{\text{SAR}}^i}{\sqrt{d_{\text{head}}}}\right), & i = 1, 2, \ldots, \text{heads} \\ \mathbf{S}_{\text{SAR}}^i = \text{Softmax}\left(\dfrac{\mathbf{Q}_{\text{SAR}}^i \mathbf{K}_{\text{OPT}}^i}{\sqrt{d_{\text{head}}}}\right). \end{cases} \tag{8}$$

The optical attention matrix $\mathbf{S}_{\text{OPT}}^i$ of each head is then multiplied by the SAR matrix $\mathbf{V}_{\text{SAR}}^i$ of each head, to obtain the amended optical feature maps of each head. Subsequently, these amended optical feature maps of all the heads are concatenated along the channel dimension, resulting in the final amended optical feature map $\mathbf{Z}_{\text{OPT}} \in \mathbb{R}^{N\times C}$. Similarly, the final amended SAR feature map $\mathbf{Z}_{\text{SAR}} \in \mathbb{R}^{N\times C}$ can be obtained in the same manner. The aforementioned process is summarized by

$$\begin{cases} \mathbf{Z}_{\text{OPT}} = \text{Concat}\left(\mathbf{S}_{\text{OPT}}^i \times \mathbf{V}_{\text{SAR}}^i\right) \\ \mathbf{Z}_{\text{SAR}} = \text{Concat}\left(\mathbf{S}_{\text{SAR}}^i \times \mathbf{V}_{\text{OPT}}^i\right) \end{cases} \quad i = 1, 2, \ldots, \text{heads}. \tag{9}$$

Next, the shapes of the final amended feature maps are transformed to $\mathbb{R}^{H\times W\times C}$ and they are added to the corresponding original input feature maps, resulting in the interactively enhanced optical feature map and SAR feature map. Finally, the two interactively enhanced modal feature maps are concatenated and their information is effectively fused through a method similar to the bottleneck structure, alleviating the gradient vanishing problem [46] caused by the network's depth and obtaining the final fused feature map $\mathbf{Fuse} \in \mathbb{R}^{H\times W\times C}$. It is worth noting that depth-wise convolutions are used to replace the normal 3×3 convolutions, maintaining module effectiveness while reducing the parameter count.

C. Decoder

To maintain the efficiency of the model, the decoder is implemented using only simple components. The specific steps are as follows. First, the multilevel fusion features $\mathbf{F}_{\text{En}}^i (1 \le i \le 4)$ from the FFM are passed through the linear layers to unify the output channel. The features are then upsampled to 1/4 of the original size and concatenated together to obtain $\mathbf{F}_{\text{up}}$. Finally, the concatenated features undergo processing through two 1×1 convolutional layers to obtain $\mathbf{F}_{\text{De}}$, which is upsampled to generate the final prediction result P. The above steps can be formulated as shown in (10). The model's accuracy can be impacted by the quantity of the branches, as detailed in Section III-D5

$$\begin{cases} \hat{F}_{\text{En}}^i &= \text{Linear}(C_{\text{in}}, C)\left(F_{\text{En}}^i\right) \\ F_{\text{up}} &= \text{Concat}\left(\text{Upsample}\left(\dfrac{H}{4} \times \dfrac{W}{4}\right)\left(\hat{F}_{\text{En}}^i\right)\right) \\ F_{\text{De}} &= \text{Conv}_{1\times1}(C, C_{\text{cls}})\left(\text{Conv}_{1\times1}(4C, C)\left(F_{\text{up}}\right)\right) \\ P &= \text{Upsample}(H \times W)(F_{\text{De}}) \end{cases} \tag{10}$$

where $\text{Linear}(C_{\text{in}}, C_{\text{cls}})(\cdot)$ and $\text{Conv}_{1\times1}(C_{\text{in}}, C_{\text{cls}})(\cdot)$ represent a linear layer and a 1×1 convolutional layer with C_{in} and C_{cls} as the input and output channels, respectively. C_{cls} is the number of prediction classes. $\text{Upsample}(H \times W)(\cdot)$D refers to upsampling the input feature to size $\mathbb{R}^{H\times W}$.

III. Experiments

A. Dataset Description

1) WHU-OPT-SAR: This dataset was built by Wuhan University and spans 51 448.56 km2 in Hubei province of China [10]. It combines 100 pairs of optical images (RGBN, where N is near-infrared) and SAR images with a size of 5556×3704, as well as the corresponding pixel-level labels. The SAR images were sourced from GaoFen-3 with a sampling resolution of 5 m. The optical images sourced from

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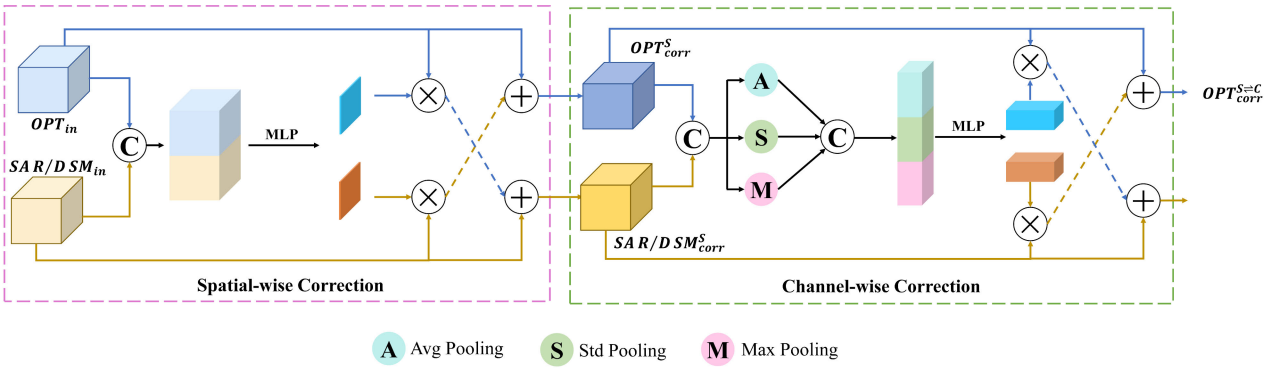


Fig. 3. FCM enhances multimodal feature extraction and fusion by adaptively correcting and filtering differences via spatial and channel correlation.

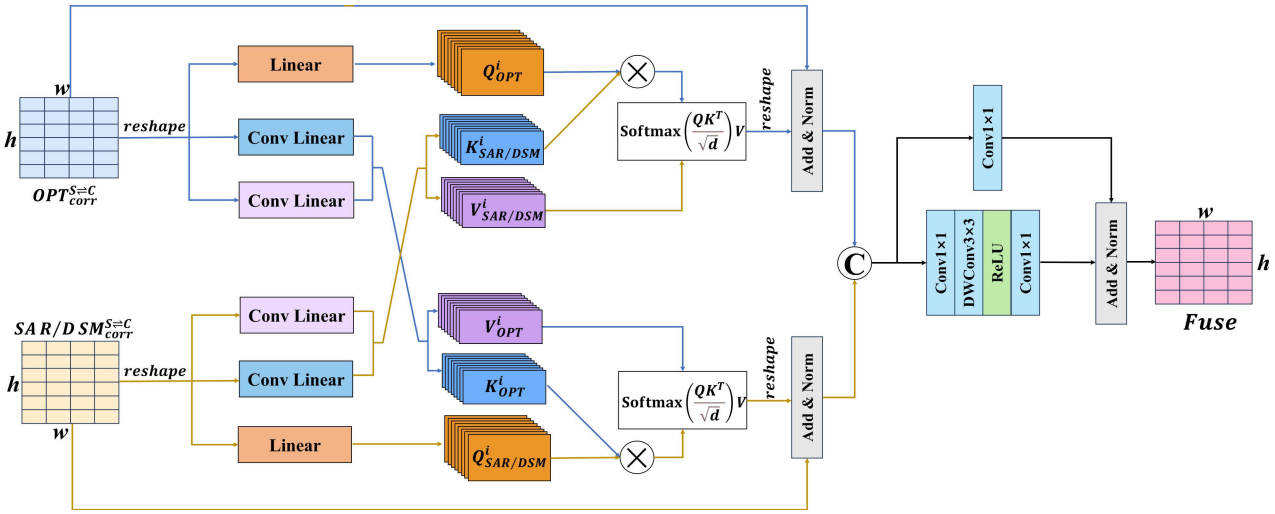


Fig. 4. FFM effectively extracts pertinent information across modalities for global interaction, comprehensively capturing the intricate nonlinear correlations within multimodal data.

GaoFen-1 (sampling resolution of 2 m) were resampled to the resolution of 5 m by bilinear interpolation. The WHU-OPT-SAR dataset contains seven land-cover categories: farmland, forest, city, village, water, road, and others. To facilitate the model training and testing, we used 80 of the images for training and 20 images for the testing, according to the hold-out strategy in the original dataset [10]. During the network design and debugging phase, we separated 20 images from the training set as the validation set. Once the network stabilized, all 80 images were used for the training. In the data preprocessing, considering the limited GPU memory, all the images were cropped into a set of subimages of size 512 × 512 without overlapping.

2) Vaihingen: The Vaihingen dataset [47] is a benchmark dataset of RSIs supplied by the International Society for Photogrammetry and Remote Sensing (ISPRS), which contains six classes, namely, impervious surface, buildings, low vegetation, trees, cars, and clutter/background. It includes 33 tiles with a spatial resolution of 9 cm, showcasing true ortho photo (TOP) images that consist of the near-infrared, red, and green (IRRG) bands. In the experiments, we also used the normalized DSM of Gerke [48]. Following previous studies [49], [50], we utilized samples with IDs 2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 27, 29, 31, 33, 35, and 38 for the testing, while the remaining 16 samples were divided into training and validation sets at different ratios in two different stages. During the network design and debugging phase, samples with IDs 1 and 13 were selected as the validation set, with the remaining samples used as the training set. After the network became stable, all 16 samples were used as the training set, and no independent validation set was used. In the data preprocessing, considering the limited GPU memory, each image was overlap-partitioned into a set of subimages sized 256 × 256 with a step size of 128 × 128.

3) Potsdam: The Potsdam dataset [51] is another benchmark dataset of RSIs provided by the ISPRS, sharing category information identical to the Vaihingen dataset. It includes 38 very fine spatial resolution TOP image tiles (5 cm), each sized 6000 × 6000 pixels. The dataset provides four bands of optical images (RGBN), along with a DSM and normalized DSM. In the experiments, following previous studies [52] and [53], we used 14 samples with IDs 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15,5_13, 5_14, 5_15, 6_13, 6_14, 6_15, and 7_13 for the testing, while the remaining 23 samples (except ID 7_10 with wrong annotations) were adjusted in two different phases. In the network design and debugging phase, samples with IDs 2_11, 4_11, 6_9, and 6_11 were incorporated into the validation set, and the remaining samples

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were used for the training. Once the network stabilized, all 23 samples were used for the training. In the data preprocessing, considering the limited GPU memory, each image was partitioned without overlapping into a set of subimages of size 512 × 512.

B. Experimental Settings

1) Implementation Details: All the experiments were conducted on an NVIDIA RTX 4090 GPU using the PyTorch framework [54]. To expedite convergence, we employed the AdamW optimizer [55] to train the model, and the weight decay was set to 0.02. The learning rate was employed with the WarmUp strategy and MultiStepLR strategy, with an initial learning rate of $5e^{-5}$. Following the WarmUp strategy, the learning rate was gradually increased to $1e^{-4}$ within a fixed epoch (5 for the WHU-OPT-SAR and Potsdam datasets, 10 for the Vaihingen dataset). To prevent the effect of random initialization parameters on the model training, a consistent random seed of 42 was used for all the experiments. The cross-entropy loss was chosen as the loss function.

In the training period, data augmentation was applied through random rotation, random flipping, and random scaling [0.5, 0.75, 1, 1.25, 1.5, 1.75]. More specifically, for the WHU-OPT-SAR dataset, the training epochs numbered 60 while the batch size was set to 6. For the Vaihingen dataset, the training epochs numbered 150 while the batch size was set to 24. For the Potsdam dataset, the training epochs numbered 50 while the batch size was set to 6. In the testing phase, overlapping sliding windows were employed to extract the small predicted patches. After obtaining the prediction results, the average value was taken for the overlapping parts, effectively correcting the minor errors at the boundaries caused by stitching, and further reducing the boundary effects. Meanwhile, to keep the inference time within an acceptable range, the overlap was set to 1/3. In addition, multiscale augmentation with a scale of [0.75, 1, 1.25] and random flipping were applied.

2) Evaluation Metrics: To validate the effectiveness of the proposed CFFormer framework, we used two main categories of evaluation metrics in the experiments. The first category of methods was used to assess the model's accuracy, i.e., the overall accuracy (OA), mean intersection over union (mIoU), Kappa coefficient (Kappa), and $F1$-score [22], [61]. The second category of methods was used to evaluate the model's scale, i.e., the assessment of complexity through the floating point operation count (FLOPs) and the number of model parameters (Params), as well as evaluating the inference speed in frames per second (FPS).

3) Comparative Models: To comprehensively validate the superiority of the proposed model on the WHU-OPT-SAR, Vaihingen, and Potsdam datasets, we compared CFFormer with 11 other SOTA methods.

a) CFFormer: The CFFormer network is a framework designed specifically for the semantic segmentation of multisource RSIs. It processes extracted optical modal features and complementary modal features through the FCM and FFM. We built three versions of the network—CFFormer-Tiny, CFFormer-Small, and CFFormer-Base—corresponding to the use of BiFormer-Tiny, BiFormer-Small, and BiFormer-Base [35] as the backbone network, respectively.

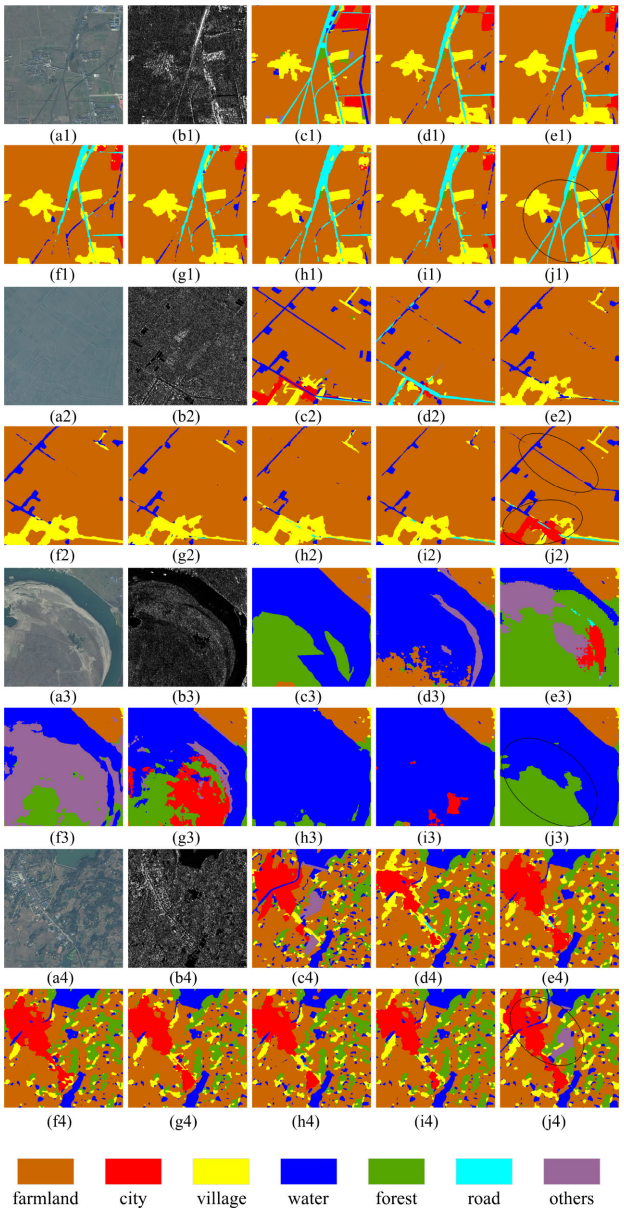


Fig. 5. Visualization of the segmentation results for the WHU-OPT-SAR dataset. (a1)–(a4) Optical image. (b1)–(b4) SAR image. (c1)–(c4) Ground truth. (d1)–(d4) V_FuseNet. (e1)–(e4) CFNet. (f1)–(f4) MCANet. (g1)–(g4) CMGFNet. (h1)–(h4) CMX. (i1)–(i4) AFNet. (j1)–(j4) Proposed-Tiny.

b) DeepLabv3+: Chen et al. [56] proposed this network. The network extends the DeepLabv3 network [29] by refining the segmentation results with a simple and effective decoder module. Furthermore, it applies depth-wise separable convolution to the atrous spatial pyramid pooling (ASPP) module, effectively enhancing the segmentation efficiency of the model.

c) SVL_3: Gerke [48] proposed this network. This method is based on stair vision library (SVL) features while simultaneously incorporating features from the normalized difference vegetation index (NDVI), saturation data, and DSM data. The classifier employs an AdaBoost-based approach, and

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TABLE I
QUANTITATIVE COMPARISON RESULTS ON THE WHU-OPT-SAR DATASET

Method	F1-score (%)							OA (%)	mIoU (%)
	Farmland	City	Village	Water	Forest	Road	Others		
DeepLabv3+(OPT) [56]	80.7	71.0	64.1	75.0	90.5	49.6	28.3	82.95	51.72
DeepLabv3+(OPT-SAR) [56]	80.5	69.8	63.0	75.8	90.4	50.6	29.8	82.87	51.68
SA-Gate [57]	79.7	71.4	64.9	73.2	90.4	54.6	22.2	81.92	51.41
V_FuseNet [24]	79.4	70.4	60.2	73.7	90.7	51.3	28.8	82.35	50.82
CFNet [28]	80.6	69.8	63.4	74.7	90.5	52.1	24.1	82.78	51.25
MCANet [10]	81.1	67.6	64.2	77.4	90.7	52.1	22.4	83.17	51.52
CMGFNet [58]	80.9	72.0	63.6	76.8	91.0	52.6	30.7	83.41	52.92
CMX [59]	81.5	70.0	65.3	74.3	91.2	56.3	33.4	83.56	53.34
AFNet [60]	81.1	71.5	64.1	77.2	91.0	54.1	31.4	83.61	53.29
Proposed-Tiny	83.0	**74.6**	67.2	79.7	91.7	61.6	41.2	85.13	57.51
Proposed-Small	83.2	74.5	67.5	80.1	91.6	61.9	41.5	85.24	57.72
Proposed-Base	**83.3**	74.5	**67.8**	**80.5**	**91.7**	**62.7**	**41.6**	**85.35**	**58.00**

a conditional random field (CRF) algorithm is introduced for postprocessing the results.

d) UFMG_4: Nogueira et al. [62] proposed this method. Based on a cascaded CNN network, this method utilizes dilated convolutions instead of regular convolutions.

e) CASIA2: Liu et al. [5] proposed this method. This network is based on the U-Net network and introduces a self-cascaded CNN module for fusing multiscale features. The network's encoder employs VGGNet and ResNet, while only IRRG data are used for the training and inference.

f) SA-Gate: Chen et al. [57] proposed this network. This network is designed with a separation and aggregation gate module, which filters and calibrates the two modal features before fusion. This design has demonstrated excellent performances in RGB-D semantic segmentation tasks.

g) V_FuseNet: Audebert et al. [24] proposed this network. This network utilizes multiscale features for loss computation, and experimental analyses are performed for early fusion and late fusion of multimodal features.

h) CFNet: Kang et al. [28] proposed this network. This network is based on a multisource RSI classification network that allows flexible adjustment of fusion positions. In addition, it incorporates a lightweight multiscale feature extraction module to fully leverage the features from different modalities.

i) MCANet: Li et al. [10] proposed this network. This network is based on a semantic segmentation framework that integrates optical and SAR images. It uses a parallel dual-encoder structure to extract optical and SAR features separately, and effectively fuses these two modalities through a cross-attention module. It is worth noting that the WHU-OPT-SAR dataset used in this article was open-sourced by the authors.

j) CMGFNet: Hosseinpour et al. [58] proposed this network. This method employs two independent encoders to extract multilevel features from high-resolution RSI and DSM images. Through a gate fusion module, it combines features from both modalities, and a top-down strategy is used to fuse the high-level features with low-level features.

k) CMX: Zhang et al. [59] proposed this network. This network is a universal cross-modal fusion architecture that adopts a dual-stream structure to independently extract features from the RGB modality and other modalities. It has achieved superior results on several multimodal datasets.

l) AFNet: Yang et al. [60] proposed this network. This network uses a multipath encoder structure that is capable of simultaneously extracting optical and DSM features. In addition, it incorporates a multipath attention fusion module for fusing features from different modalities and a refined attention fusion module for combining shallow and deep features. The network achieved SOTA results on the Vaihingen dataset and Potsdam dataset at the time of publication.

m) BEDSN: Li et al. [63] proposed this network. This network is a new boundary-enhanced dual-flow network (BEDSN) in which an edge-detection branch stream with a composite loss function (EDBS) is introduced to compensate for the boundary loss in the semantically segmented branch stream (SSBS). EDBS and SSBS are integrated through highly coupled encoders and feature extractors. A lightweight multilevel information fusion module guided by the channel attention mechanism is designed to reuse intermediate boundary information effectively. The multiscale feature extraction module and the mixed attribute convolution module are used to enhance the SSBS for the aggregation of multiscale context information.

C. Experimental Results

Results Obtained on the WHU-OPT-SAR Dataset: WHU-OPT-SAR is a large-scale land-use classification dataset of multisource RSIs, covering an area of about 50 000 km2. In this study, we compared the three versions of the proposed model with several other advanced multimodal semantic segmentation networks on the official WHU-OPT-SAR test set. In addition, we conducted two sets of experiments using the DeepLabv3 + network: one with optical images as the input, named DeepLabv3 + (OPT), and another with a concatenation of optical and SAR images as the input, named DeepLabv3 + (OPT-SAR). As illustrated in Table I, the results of DeepLabv3 + (OPT) and DeepLabv3 + (OPT-SAR) suggest that simple fusion not only fails to effectively utilize the multimodal information but may also even degrade the model's performance to some extent. Furthermore, the three versions of the proposed model exhibit a significant advantage across all the metrics. Taking the example of the CFFormer-Base network, which adopts BiFormer-Base [39] as the backbone network, it shows improvements in mIoU

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TABLE II
QUANTITATIVE COMPARISON RESULTS ON THE VAIHINGEN DATASET

Method	F1-score(%)					mIoU(%)	OA(%)
	Imp.Surf.	Building	Low Veg.	Tree	Car		
SVL_3 [48]	86.6	91.0	77.0	85.0	55.6	-	84.80
UFMG_4 [62]	91.1	94.5	82.9	88.8	81.3	-	89.40
CASIA2 [5]	93.2	96.0	84.7	89.9	86.7	-	91.10
BEDSN [63]	92.3	94.7	83.7	89.2	86.3	80.82	90.14
SA-Gate [57]	92.6	96.0	83.5	89.5	86.9	81.62	90.61
V_FuseNet [24]	92.4	95.4	83.1	89.4	86.8	81.14	90.30
CFNet [28]	92.2	95.8	83.2	84.8	81.6	80.43	90.29
MCANet [10]	92.1	95.6	83.0	89.0	87.5	81.19	90.18
CMGFNet [58]	92.3	95.8	83.4	89.2	85.3	80.81	90.39
CMX [59]	92.4	96.3	84.1	89.6	86.5	81.76	90.80
AFNet [60]	92.2	96.0	83.4	89.4	85.1	80.85	90.40
Proposed-Tiny	93.1	**96.6**	85.8	90.7	90.0	84.06	91.72
Proposed-Small	**93.7**	96.5	**86.0**	**90.7**	**90.0**	**84.31**	**91.88**
Proposed-Base	92.4	96.0	85.9	87.9	87.5	81.72	90.74

and OA over the previously best-performing AFNet network of 4.71% and 1.74%, respectively, while maintaining an advantage in $F1$-score for each category. Compared with the CFNet and MCANet networks specifically designed for handling optical and SAR images, CFFormer-Base achieves a notable improvement of 6.75% and 6.48% in mIoU, demonstrating the effectiveness of the proposed method. In addition, when dealing with challenging segmentation tasks for elongated categories such as roads and small objects such as the others category, the proposed model shows an impressive performance. Specifically, the $F1$-score for the road category and others category with the CFFormer-Base network is 62.7% and 41.6%, respectively, surpassing the other methods by an average of 10.1% and 13.7%, effectively improving the results of RSI segmentation. Furthermore, the visualization results in four different regions of the WHU-OPT-SAR test set (see Fig. 5) also demonstrate the effectiveness of the proposed method. In the first and second regions of the experiments, CFFormer establishes long-range dependencies between distant pixels by using the transformer-based backbone networks, leveraging global contextual information to significantly improve the continuity of road [see Fig. 5(j1)] and water [see Fig. 5(j2)] segmentation in complex scenes. In the last two regions of the experiments, the other advanced methods show large-scale confusion, while the proposed approach eliminates noise and the differences in the cross-modal data through the FCM and further extracts the complementary features by using the FFM to effectively alleviate this issue. More results are presented in Fig. 6.

Results Obtained on the Vaihingen Dataset: The Vaihingen dataset is widely used in RSI segmentation tasks. Up to now, there have been a number of models designed specifically for this dataset that have achieved a high accuracy. To comprehensively compare the proposed method with the other methods, we selected three models that have performed well on the Vaihingen and Potsdam dataset leaderboards—SVL_3, UFMG_4, and CASIA2—which are unimodal segmentation networks that use only optical RSIs. In addition, seven advanced multimodal RSI segmentation networks were used to compare the segmentation results in multimodal scenarios. The experimental results are listed in Table II, where it can be seen that the proposed network significantly outperforms the other methods in segmentation accuracy. The mIoU and OA for the tiny version and small version and base version

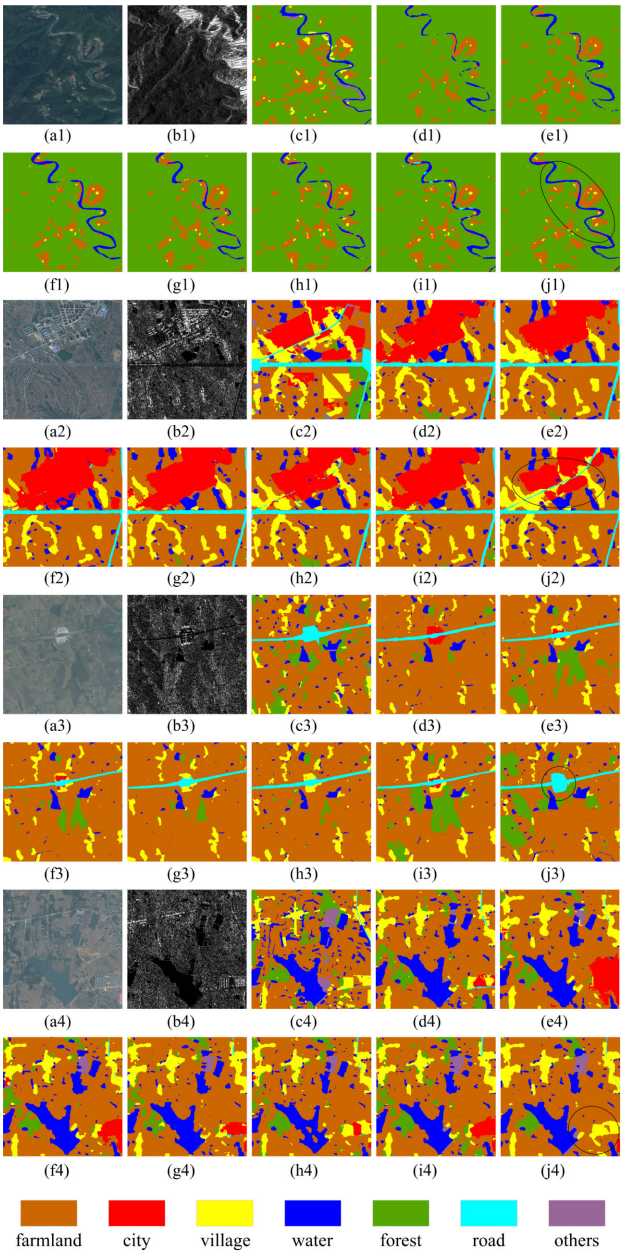


Fig. 6. Visualization of the segmentation results for the WHU-OPT-SAR dataset. (a1)–(a4) Optical image. (b1)–(b4) SAR image. (c1)–(c4) Ground truth. (d1)–(d4) V_FuseNet. (e1)–(e4) CFNet. (f1)–(f4) MCANet. (g1)–(g4) CMGFNet. (h1)–(h4) CMX. (i1)–(i4) AFNet. (j1)–(j4) Proposed.

are 84.06%/91.72%, 84.31%/91.88%, and 81.72%/90.74%, respectively. Specifically, the OA of CFFormer-Small is improved by 0.78% when compared to CASIA2, which performs the best among the unimodal networks, and by 1.08% when compared to CMX, which performs the best among the multimodal networks. Furthermore, it can be observed that, although the accuracy of most of the multimodal segmentation networks is higher than that of the unimodal segmentation networks, the opposite also exists. This is because there can be serious noise and differences between the features of the different modalities. Without effective handling, this can lead to mutual interference between the different modalities, thereby reducing the segmentation accuracy. The mIoU of the

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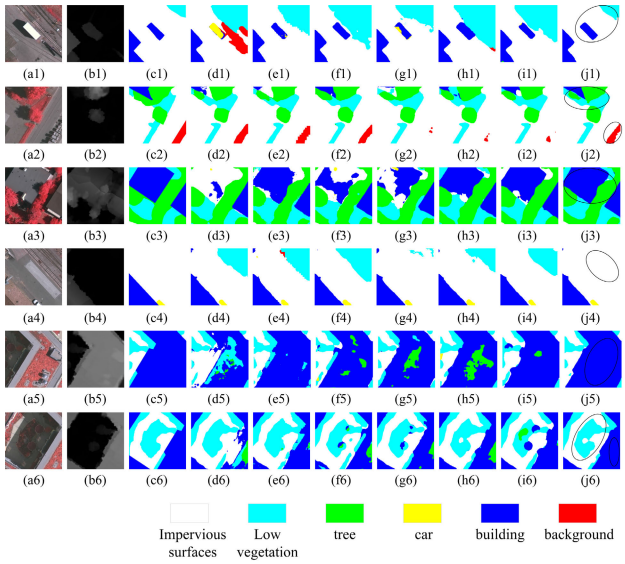


Fig. 7. Visualization of the segmentation results for the Vaihingen dataset. (a1)–(a6) Optical image. (b1)–(b6) DSM image. (c1)–(c6) Ground truth. (d1)–(d6) V_FuseNet. (e1)–(e6) CFNet. (f1)–(f6) MCANet. (g1)–(g6) CMGFNet. (h1)–(h6) CMX. (i1)–(i6) AFNet. (j1)–(j6) Proposed-Small.

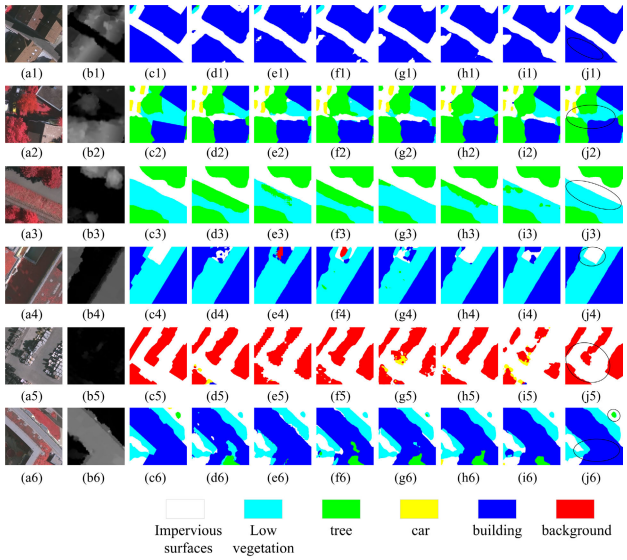


Fig. 8. Visualization of the segmentation results for the Vaihingen dataset. (a1)–(a6) Optical image. (b1)–(b6) DSM image. (c1)–(c6) Ground truth. (d1)–(d6) V_FuseNet. (e1)–(e6) CFNet. (f1)–(f6) MCANet. (g1)–(g6) CMGFNet. (h1)–(h6) CMX. (i1)–(i6) AFNet. (j1)–(j6) Proposed.

Proposed-Base is lower compared to the Proposed-Tiny and Proposed-Small. This is because, compared with the WHU-OPT-SAR dataset and the Potsdam dataset, the Vaihingen dataset contains fewer images. As a variant structure based on variants with different structures. For datasets with fewer samples, the deeper the model structure is, the more likely it is to cause overfitting on the test set. Through our several experiments, we have found that the network depth of Proposed-Tiny and Proposed-Small is more suitable for the Vaihingen dataset. The proposed method reduces the differences and noise between optical images and DSM images through the FCM and captures their complementary features using the cross-attention mechanism, achieving superior results. The visual comparison of the segmentation results in Fig. 7 indicates that the segmentation map of CFFormer is the most detailed and closest to the ground truth. In contrast, the other advanced networks in this experiment exhibit different degrees of mutual misclassification between impervious surfaces, low vegetation, buildings, and trees, and their segmentation results also show coarser features [see Fig. 7(g3), (d5), and (d6)]. The proposed network fully extracts and utilizes the global information provided by the transformer-based encoder and multiscale information provided by the decoder, combined with auxiliary information from the DSM, achieving the best segmentation results. More results are presented in Fig. 8.

Results Obtained on the Potsdam Dataset: For a comprehensive evaluation, we further conducted experiments on the Potsdam dataset. The results are listed in Table III, where all three versions of the proposed models lead in all the metrics. The OA of CFFormer-Tiny is 15.19% and 4.49% higher than those of the unimodal segmentation networks SVL_3 and UFMG_4, respectively. Compared to V_FuseNet, CMGFNet, and AFNet, which are specifically designed for this dataset, CFFormer-Base improves the mIoU and OA by 5.49%/2.99%/3.24% and 3.52%/1.8%/1.86%, respectively. For the dense small object categories such as cars, the proposed method achieves a 97% $F1$-score, which is an average improvement of 5.07% over the other methods. This fully demonstrates that the proposed method not only performs excellently in overall performance, but also exhibits significant superiority in handling small targets. In addition, we also provide the segmentation results in Fig. 9 for the CFFormer-Base network and the other advanced networks to further demonstrate the superior performance of the proposed network. Consistent with its performance on the Vaihingen dataset, CFFormer accurately segments the different objects in the various challenging scenarios, including objects occluded by trees and buildings (first, second, and fourth sets of experiments) and objects with similar spectral characteristics (third set of experiments). These results indicate that, compared to the other multimodal segmentation networks, the FCM and FFM in the proposed approach can comprehensively utilize the complementary information from optical and elevation features for collaborative scene understanding to obtain the most accurate segmentation results. More results are presented in Fig. 10.

TABLE III
QUANTITATIVE COMPARISON RESULTS ON THE POTSDAM DATASET

Method	F1-score (%)					mIoU (%)	OA (%)
	Imp.Surf.	Building	Low Veg.	Tree	Car		
SVL_3 [48]	84.0	89.8	72.0	89.0	69.8	-	77.2
UFMG_4 [62]	90.8	95.6	84.4	84.3	92.4	-	87.90
CASIA2 [5]	93.3	97.0	87.7	88.4	96.2	-	91.10
BEDSN [63]	91.8	95.6	85.9	86.7	95.0	-	89.17
SA-Gate [57]	92.5	96.2	87.2	87.9	93.7	84.49	89.97
V_FuseNet [24]	91.6	95.9	85.8	86.3	93.6	83.13	89.12
CFNet [28]	92.8	97.1	86.1	87.7	95.0	85.01	90.06
MCANet [10]	92.9	96.9	87.4	88.1	95.3	85.63	90.62
CMGFNet [58]	92.8	97.2	87.6	88.3	94.8	85.63	90.84
CMX [59]	93.0	97.3	88.1	89.5	94.0	85.64	91.08
AFNet [60]	93.0	97.2	87.3	87.9	94.5	85.38	90.78
Proposed-Tiny	94.5	97.9	89.2	89.8	96.5	88.14	92.39
Proposed-Small	94.3	**97.9**	**89.6**	**90.5**	97.0	88.62	92.52
Proposed-Base	**94.7**	97.7	89.5	90.4	**97.0**	**88.62**	**92.64**

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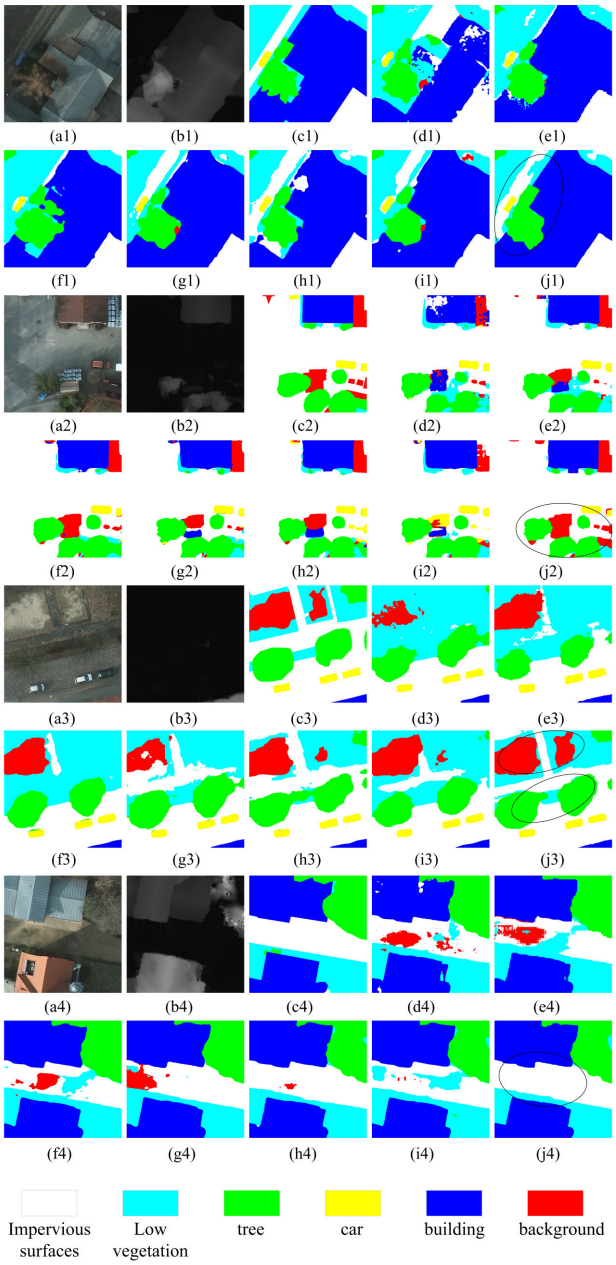


Fig. 9. Visualization of the segmentation results for the Potsdam dataset. (a1)–(a4) Optical image. (b1)–(b4) DSM image. (c1)–(c4) Ground truth. (d1)–(d4) V_FuseNet. (e1)–(e4) CFNet. (f1)–(f4) MCANet. (g1)–(g4) CMGFNet. (h1)–(h4) CMX. (i1)–(i4) AFNet. (j1)–(j4) Proposed-Base.

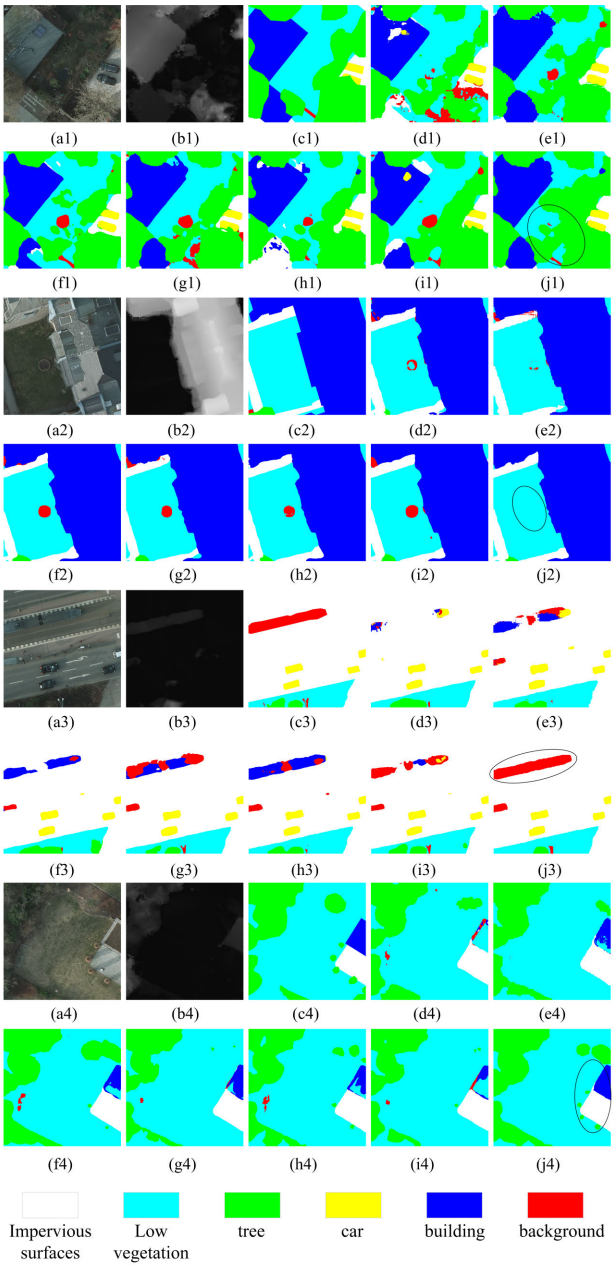


Fig. 10. Visualization of the segmentation results for the Potsdam dataset. (a1)–(a4) Optical image. (b1)–(b4) DSM image. (c1)–(c4) Ground truth. (d1)–(d4) V_FuseNet. (e1)–(e4) CFNet. (f1)–(f4) MCANet. (g1)–(g4) CMGFNet. (h1)–(h4) CMX. (i1)–(i4) AFNet. (j1)–(j4) Proposed.

TABLE IV

ABLATION STUDY FOR THE FCM/FFM ON THE WHU-OPT-SAR DATASET

FCM	FFM	mIoU(%)	OA(%)
		55.26	84.31
✓		56.28	84.62
	✓	56.01	84.63
✓	✓	**57.51**	**85.13**

TABLE V

ABLATION STUDY FOR THE FCM VARIANTS ON THE WHU-OPT-SAR DATASET

FCM	mIoU(%)	OA(%)
Channel->Spatial	57.27	85.11
Spatial->Channel	**57.51**	**85.13**

Overall Analyses of Datasets: The types of the three datasets are not consistent. The WHU-OPT-SAR dataset (see Table I) includes SAR and optical images, while the Vaihingen and Potsdam datasets (see Tables II and III) contain optical and DSM images. The difference among the three versions of the proposed methods lies in the network's depth. For the Proposed-Tiny version, the model has the shallowest layers. Based on our experimental results on the three datasets, it can be observed that, compared to other categories, building or city categories with more regular shapes often achieve

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TABLE VI
ABLATION STUDY FOR THE FFM VARIANTS ON THE WHU-OPT-SAR DATASET

FFM	mIoU(%)	OA(%)
Add	56.86	84.97
Concat	57.23	85.03
CrossAttn(proposed)	**57.51**	**85.13**

better segmentation when the model has shallower layers. This indicates that the shallower layers of the network focus more on local texture information [64], [65]. Therefore, for Proposed-Tiny, the segmentation of large land cover categories with regular block-like shapes (such as buildings or cities) is usually more effective. In the Potsdam dataset, the data sources used are optical and DSM images. In optical RSIs, it is difficult to distinguish between trees and low vegetation with similar color and texture information. The elevation information provided by DSM can compensate for the shortcomings of optical RSIs. As the depth of the model increases, the model can capture more information about both modalities. As a result, the trees and low vegetation categories in the Proposed-Small version have better accuracy than the Proposed-Tiny version. However, models with deeper structures may dilute the height information provided by the DSM [66], [67], which leads to confusion during feature fusion. That is the reason why the results of trees and low vegetation in the Proposed-Base version are less accurate than the Proposed-Small version.

D. Ablation Study

To evaluate the effectiveness of each module in the proposed method, we carried out a series of ablation experiments, which are elaborated in this section. All the experiments involved training and testing on the WHU-OPT-SAR dataset. Unless otherwise specified, we used BiFormer-Tiny as the backbone network for the ablation experiments.

1) Effectiveness of the FCM and FFM: As shown in Fig. 2, we embedded the FCM and FFM in the encoder–decoder structure to correct and fuse two sets of features from different modalities. To evaluate the impact of these two modules on the segmentation accuracy of the CFFormer framework, we conducted ablation experiments on the test set. In the experiments, "✓" indicates the inclusion of the corresponding module. If the FCM was ablated, the features extracted from the two branches were not corrected. If the FFM was not used, the extracted multimodal features were simply summed and then input into the decoder. The specific results are listed in Table IV. Compared to the baseline, adding the FCM increases the mIoU and OA by 1.02% and 0.31%, respectively, while adding the FFM increases the mIoU by 0.75%. When both the FCM and FFM are used, the effect is maximized, with the mIoU and OA increasing by 2.25% and 0.82%, respectively, achieving the effect of "1 + 1 > 2." This is because there are significant differences and noise between the features of the different modalities. Only by first reducing the mutual interference through feature correction can better interaction and fusion be achieved, leading to superior segmentation results.

2) Feature Correction First Versus Channel-wise Feature Correction First: To further explore the effect of the FCM on the CFFormer, we conducted ablation experiments on the sequence of spatial-wise feature correction and channel-wise feature correction. "Channel -> Spatial" represents performing the channel-wise correction first and then the spatial-wise correction, while "Spatial -> Channel" is the opposite. As shown in Table V, there is a 0.24% improvement in mIoU when using the spatial-wise feature correction first versus using the channel-wise feature correction first. We attribute this to the fact that the intuitive local appearance differences between the different modalities are more pronounced than the abstract global semantic differences. Therefore, achieving the best experimental results involves first eliminating the large local spatial differences through spatial-wise correction and then eliminating the slightly smaller global semantic differences through channel-wise correction.

3) Ablation Study With FFM Variants: Similarly, to further study the effect of the FFM on CFFormer, we conducted ablation experiments on the way of information fusion between different modalities. "Add" and "Concat" represent fusing the corrected features directly by addition and concatenation without using the cross-attention mechanism, and "CrossAttn" represents using the cross-attention mechanism designed in this study. The experimental results in Table VI show that, when using the cross-attention mechanism, the best segmentation accuracy is achieved, with the mIoU and OA reaching 57.51% and 85.13%, respectively. This further demonstrates that, by transforming the corrected features into sequences and globally interacting the information between different modalities from a sequence-to-sequence perspective, complementary relationships can be effectively mined, thereby improving the model's performance. It is worth noting that the "Concat" method achieves a 0.37% increase in mIoU when compared to the "Add" method. This is because the sparsity between the features of different modalities is weak, and most positions in the data contain valid information. In this case, using concatenation provides more comprehensive and richer fusion features.

4) Effectiveness of Different Types of Encoders: Currently, remote sensing-based segmentation networks commonly apply a CNN as the encoder. This is because CNNs can effectively capture the local correlations in the input images through convolution operations. However, due to the local nature of the convolution operators, CNNs have significant limitations in modeling the global relationships in images. In contrast, transformers, through the self-attention mechanisms, excel at capturing the global information in RSIs while establishing the based BiFormer-Tiny network, BiFormer-Small network, long-range dependencies in information-dense multimodal BiFormer-Base network and a hybrid SMT-Tiny network [57] as encoders in the proposed framework on the WHU-OPT-SAR dataset. The experimental results listed in Table VII show that the transformer model achieves the best accuracies in terms of the mIoU and OA, reaching 58.0% and 85.3%, respectively. Compared to the CNN model, it improves

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TABLE VII
RESULTS FOR DIFFERENT TYPES OF ENCODERS ON THE WHU-OPT-SAR DATASET. THE COMPLEXITY AND SPEED WERE MEASURED BY THE 512 × 512 INPUT OF OPT AND SAR DATA ON AN NVIDIA RTX 4090 GPU

Backbone	Encoder	Parameters(M) ↓	FLOPs(G) ↓	Speed(FPS)	mIoU(%)	OA(%)
ResNet-18 [46]	CNN	**39.4**	**98.0**	**60.6**	54.3	83.8
SMT-Tiny [68]	CNN-transformer	48.4	111.9	23.6	56.3	84.6
Proposed-Tiny	Transformer	51.7	105.9	21.1	57.5	85.1
Proposed-Small	Transformer	76.4	150.4	13.53	57.7	85.2
Proposed-Base	Transformer	168.8	272.1	14.06	**58.0**	**85.3**

TABLE VIII
ABLATION STUDY FOR THE NUMBER OF DECODER LAYERS ON THE WHU-OPT-SAR DATASET

Number	F1-score (%)						mIoU(%)	Δa(%)
	City	Village	Water	Forest	Road	Others		
1	67.0	62.1	74.8	90.7	47.6	17.1	49.34	-
2	71.0	64.8	77.8	91.2	56.4	31.4	53.93	4.59↑
3	73.8	66.6	**79.9**	91.5	60.3	38.8	56.67	7.33↑
4(proposed)	**74.6**	**67.2**	79.7	**91.7**	**61.6**	**41.2**	**57.51**	**8.17↑**
5	72.7	65.8	77.4	91.0	57.9	36.1	54.97	5.63↑

the mIoU and OA by 3.2% and 1.3%, respectively, which effectively proves the importance of global information and long-range dependency modeling for multimodal RSI segmentation tasks. Meanwhile, the CNN model achieves advantages in model complexity and inference speed. Especially, the inference speed of CNN is 4.4×, 4.3×, 2.8×, and 2.5× faster than the three transformer models and the hybrid CNN-transformer, respectively. CNN-transformer performs at a moderate level in almost all the metrics, with a slightly improved accuracy when compared to the CNN model, but with significant differences in speed and complexity. Therefore, considering the characteristics of multimodal RSIs, we finally chose the transformer model as the encoder of the proposed method, while we believe that CNN models still hold significant potential, especially for real-time segmentation tasks.

5) Effectiveness of the Number of Decoder Layers: The encoder–decoder architecture, which is known for its outstanding performance and efficient flexibility, has been widely applied in semantic segmentation tasks. In this study, we fully leveraged the characteristics of this architecture by using the downsampled features at each stage of the backbone network as inputs to the corresponding FCM. After obtaining fused features at different scales, we employed multiple convolutional layers and upsampling operations to restore the spatial resolution of these features to have the same dimensions. During the experiments, we observed that features from a single layer, or a few layers, may lack sufficient contextual information, leading to the loss of critical details when the model comprehends the overall image context. Nevertheless, excessively deep architectures not only escalate computational expenses but may also result in overfitting during the training phase. Our experimental results suggest that the four-layer structure is the most reliable and practical. As shown in Table VIII, the comprehensive use of features from four scales yields the best results, with a mIoU of 57.51%, which is an improvement of 8.17% compared to using features from a single layer. Moreover, the $F1$-score for the other and road categories experienced significant increases of 240% (from 17.1% to 41.2%) and 130% (from 47.6% to 61.6%), respectively. Before the fourth layer, the accuracy of the network increased the importance of multiscale features in RSI segmentation tasks with each additional feature layer, which further demonstrates.

TABLE IX
ABLATION STUDY OF STANDARD DEVIATION POOLING OF FCM MODULE ON WHU-OPT-SAR DATASET

Avg+Max	Std	mIoU(%)	OA(%)
✓		56.69	84.74
✓	✓	**57.51**	**85.13**

6) Effectiveness of Standard Deviation Pooling in FCM Modules: As illustrated in Fig. 3, we have integrated global average pooling, global maximum pooling, and global standard deviation pooling within the channel-wise correction mechanism of the FCM module. To gauge the influence of standard deviation pooling on the segmentation accuracy within the CFFormer architecture, we performed ablation studies using the test dataset. In these experiments, the presence of a module is indicated by a "✓." When global standard deviation pooling is engaged, it operates alongside global average and maximum pooling. The comprehensive results are detailed in Table IX. The introduction of global standard deviation pooling, in contrast to using global average and global maximum pooling alone, boosts the mIoU by 0.82% and the OA by 0.39%. Global standard deviation pooling enables the model to focus on feature changes and distribution information. It allows our model to concentrate on more meaningful features and adapt to different types and data distributions, resulting in more refined segmentation results.

E. Efficiency Analysis

Model accuracy and model complexity are critical factors in assessing the overall performance of a network. Therefore, we used six metrics—mIoU, OA, m$F1$-Score, Kappa, number

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TABLE X
EFFICIENCY RESULTS ON THE WHU-OPT-SAR DATASET. THE COMPLEXITY WAS MEASURED BY THE 512 × 512 INPUT OF OPT AND SAR DATA ON AN NVIDIA RTX 4090 GPU

Method	Parameters (M)↓	FLOPs (G)↓	mIoU (%)	OA (%)	mF1-Score (%)	Kappa (%)
SA-Gate [57]	110.9	330.1	51.41	81.92	65.2	72.8
V_FuseNet [24]	66.7	692.2	50.82	82.35	64.9	72.9
CFNet [28]	**4.3**	**78.9**	51.25	82.78	65.0	73.7
MCANet [10]	189.9	3035	51.52	83.41	66.8	74.7
CMGFNet [58]	85.2	312.6	52.92	83.41	65.1	74.5
CMX [59]	77.4	128.8	53.34	83.56	67.4	74.8
AFNet [60]	71.3	260.85	53.29	83.61	67.2	74.9
Proposed-Tiny	51.7	105.9	57.51	85.13	71.3	77.3
Proposed-Small	76.4	150.4	57.72	85.24	71.5	77.5
Proposed-Base	168.8	272.1	**58.00**	**85.35**	**71.7**	**77.6**

TABLE XI
ABLATION STUDY OF DIFFERENT MODALITIES ON THE WHU-OPT-SAR DATASET

	Data	mIoU(%)	OA(%)
Unimodal	OPT	54.68	84.12
Multimodal	OPT+SAR	**57.51**	**85.13**

of parameters, and FLOPs—to compare the proposed method on the WHU-OPT-SAR dataset based on model accuracy and complexity. In the experiments, two BiFormer-Tiny backbone networks were employed to extract optical and SAR features, and all other model methods also used two branches for feature extraction. The training and testing were conducted using the WHU-OPT-SAR dataset. The specific results are presented in Table X. It can be observed that the computational cost of the Proposed-Tiny is lower than those of many models, being only higher than that of CFNet (see Table X). Meanwhile, Proposed-Tiny has improved in all accuracy metrics, with an increase of 6.26% in mIoU, 2.35% in OA, 6.3% in mF1-Score, and 3.6% in Kappa. Our Proposed-Tiny model achieves higher accuracy while maintaining a small number of parameters and a low amount of computation. Our Proposed-Small model and Proposed-Base model achieve higher accuracy while the number of parameters is close to those of other models. The experimental results indicate that increased parameters do not always positively correlate with improved accuracy. Moreover, a good architecture and feature fusion strategy are more important than increasing parameters in image segmentation performance.

IV. QUALITATIVE ANALYSIS AND DISCUSSION

A. Unimodal and Multimodal Results

The core advantage of CFFormer is that it can fully extract and utilize the complementary information between different modalities to obtain more accurate segmentation results. To demonstrate this, we show the segmentation results of the BiFormer-Tiny network using only optical images and the CFFormer-Tiny network jointly using optical images and SAR or DSM images in Fig. 11.

In the WHU-OPT-SAR dataset, as shown in Fig. 11(a), the BiFormer-Tiny network mistakenly segments the city into the village and almost fails to recognize the water near the forest due to the difficulty in distinguishing spectrally

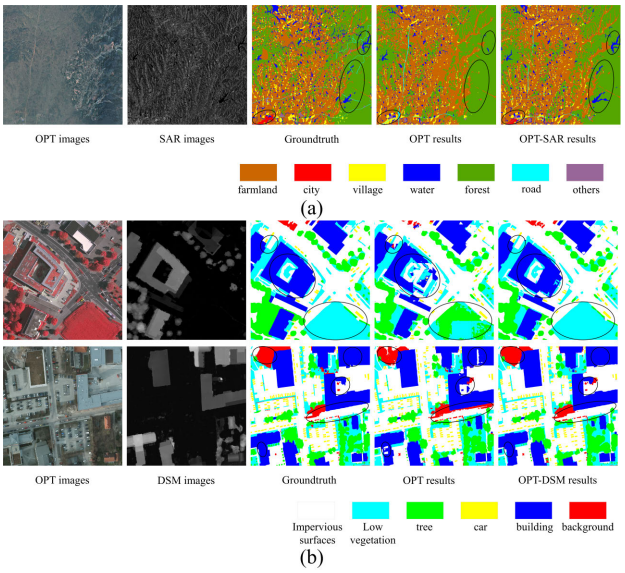


Fig. 11. Visualization of the semantic segmentation results for OPT-only and for the OPT-SAR/DSM methods. We applied Biformer-Tiny for the OPT-only segmentation and the proposed method with the same backbone of Biformer-Tiny for the OPT-SAR/DSM segmentation. (a) Visualization comparison on the WHU-OPT-SAR dataset. (b) Visualization comparison on the Vaihingen (top) and Potsdam (bottom) datasets.

similar categories such as city from village and water from forest. However, in SAR images, water presents darker features because its backscattering coefficient is low due to its relatively smooth surface, while the city presents bright features due to the presence of artificial structures such as buildings, which make the surface relatively rough and capable of inducing strong backscattering. The proposed CFFormer-Tiny makes full use of these characteristics to provide good complementary information for the optical images and realizes the accurate segmentation of the confusing categories mentioned above. In the Vaihingen and Potsdam datasets, as shown in Fig. 11(b), tree and low vegetation present highly similar appearances and contours in the optical images, causing the BiFormer-Tiny network, which relies only on the optical images, to misidentify low vegetation as trees. The proposed method successfully solves this difficulty with the help of the elevation difference information provided by the DSM. In addition, CFFormer-Tiny is also more accurate than BiFormer-Tiny in edge-detailed segmentation. In conclusion,

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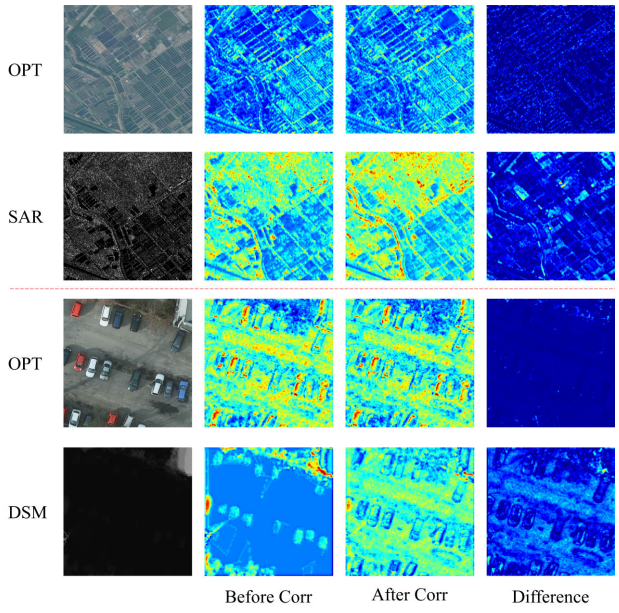


Fig. 12. Visualization of the features extracted in layer 1, their corrected counterparts, and the difference maps between them.

the experimental results fully demonstrate the powerful performance of CFFormer in the multimodal remote sensing semantic segmentation task. Compared to unimodal networks, CFFormer also exhibits stronger robustness and a superior generalization performance. We have utilized a multimodal structure to design networks integrating information from different modalities. To evaluate the effectiveness of this architectural design, we conducted ablation studies on the test dataset. Our ablation studies included single-modal networks using only optical images, and multimodal networks using both optical and SAR images. The detailed results are presented in Table XI. The segmentation result achieved with the multimodal structure surpasses those only using optical images by 2.83%. This strongly validates the effectiveness of the multimodal structure: the integration of multisource images promotes efficient information exchange through interaction, leading to better fusion and complementarity, and ultimately enhancing the accuracy of segmentation.

B. Feature Correction

To better understand the working principles of the FCM, we visualize in Fig. 12 the features extracted from layer 1 before correction, the features after passing through the FCM, as well as the difference maps between them. It is obvious that, after cross-modal feature correction, the feature maps in the different modalities are corrected and enhanced accordingly, while the intermodal disparities are significantly improved.

The optical modality transfers the spectral and textural information to the SAR and DSM modalities, while the SAR and DSM modalities locate the boundaries of the optical modality more precisely. In the second row, the uncorrected SAR modality displays irregular chunks in the farmland area, whereas, after correction, the farmland contours become clear, showing regularly distributed long strips. A similar situation is even more evident in the DSM modality, where the car and the ground are almost indistinguishable in the precorrection features, and after injecting the complementary information from the optical image, the contour of the car and the position of the ground are clearly visible. In addition, by observing the difference map of the optical features, it can be found that the boundary information of the optical features is further enhanced after correction, while some unnecessary noise is filtered out. In summary, the proposed FCM can effectively leverage the complementary information between multimodal data to eliminate the differences and noise between them, so as to realize more accurate semantic understanding.

V. Conclusion

In this article, we have proposed a universal CFFormer for the semantic segmentation of joint optical RSIs and other remote sensing data, such as SAR and DSM data. In terms of the differences and interference noise between multimodal RSIs, we designed the FCM to correct the features from the different modalities in both the spatial and channel dimensions, so that the different modalities can pay more attention to the complementary information between each other. To fully exploit the complementarity between multimodal data, we proposed the FFM, which utilizes a multihead cross-attention mechanism to interact and fuse features of the same scale from two modalities. Extensive benchmark experiments and ablation studies conducted on the WHU-OPT-SAR, Vaihingen, and Potsdam datasets demonstrated the effectiveness and robustness of the proposed approach for multisource remote sensing applications. Meanwhile, the visualization results indicated that CFFormer can effectively identify spectrally similar objects in optical images using the complementary information provided by SAR and DSM data, resulting in more accurate segmentation. However, the universal module that we presented in this study was intended to harness the complementarity of different modal information regardless of the imaging mechanism. In the future, we will design a model by considering the physical mechanisms of different images to increase its interpretability.

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